

Department of Computer Engineering

**ZEAL COLLEGE OF ENGINEERING & RESEARCH,
NARHE, PUNE**

Department of Computer Engineering



Department of Computer Engineering

COIN201: Internship

ACADEMIC YEAR 2025-26

“Web Application :- Dinex”

_____ (Students Name) _____

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Internship Place Details

Activities/Scope	Only for School and College ERP (Enterprise Resource Planning) Software and Technical Workshops for Entrepreneurship Development
Objective of Study	● Learning web development(HTML,CSS) ● Improving Python programming skills ● Gaining experience with database management ● Understanding software development life cycle ● Building a portfolio ● Building A Responsive Website ● Gaining Industry level Knowledge
Supervisor Details (Name,Designation,Company Name,Email_id,Contact number)	

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Introduction

The Dinex Web Application is a centralized, web-based platform developed to simplify the process of restaurant billing and management. In the modern food service industry, restaurants and cafes often struggle with manual billing, calculating totals, generating receipts, and keeping track of daily sales efficiently. The traditional method of maintaining paper bills is time-consuming, prone to calculation errors, and lacks proper financial organization.

To overcome these challenges, our team developed the Dinex system (accessible at <http://dinex.free.je/>), which allows restaurant staff and administrators to generate bills, manage menu prices, and track revenue through an easy-to-use interface. The system stores all financial and transactional information securely in a MySQL database, making it easier for administrators to maintain and retrieve records.

The application includes separate modules for secure staff login, adding items to a customer's bill, calculating taxes, generating final invoices, and an administrator panel. The administrator has the authority to manage the system, update menu pricing, and monitor sales. The project is developed using HTML, CSS, JavaScript, PHP, MySQL, and XAMPP, while Git and GitHub were used for version control and collaborative development.

Title:-

Dinex - Restaurant Billing & Management System

Problem Statement:-

Restaurants frequently face delays and errors due to manual billing processes. Calculating order totals, applying taxes, generating physical receipts, and tracking daily sales by hand is inefficient and can lead to financial discrepancies. Therefore, there is a need for a secure, centralized, and web-based billing platform that enables staff to generate quick, accurate invoices while allowing administrators to streamline financial records.

Objective:-

- Develop a secure Restaurant Billing Management System.
- Digitize the invoicing and receipt generation process.
- Reduce manual calculation errors and paperwork.
- Enable seamless administrator oversight of sales data.
- Provide secure staff authentication.
- Maintain centralized financial and sales records.

Motivation/Scope & Rationale of the Study

Motivation :-

The motivation behind developing the Dinex Web Application was to address the common operational bottlenecks faced by restaurant staff at the checkout counter. We wanted to create a secure and user-friendly digital platform that simplifies calculating totals, generating bills, and tracking daily revenue. This project also gave us an opportunity to apply our knowledge of full-stack web development (HTML, CSS, PHP, MySQL) in a real-world application while improving our teamwork and problem-solving skills. Our aim was to develop a system that saves time, reduces financial errors, and provides an efficient solution for restaurant management.

Scope & Rationale of the Study:-

Scope

- **Staff/Admin Registration & Login**
- **Menu & Price Management**
- **Digital Billing & Invoice Generation**
- **Tax Calculation**
- **Admin Dashboard for Sales Tracking**
- **Database Management**
- **Responsive User Interface**
- **Secure Authentication**

Rationale

The project provides a centralized platform that simplifies restaurant billing while ensuring financial security and transparency. It reduces manual calculation work, speeds up the checkout process, and gives administrators clear insights into daily operations.

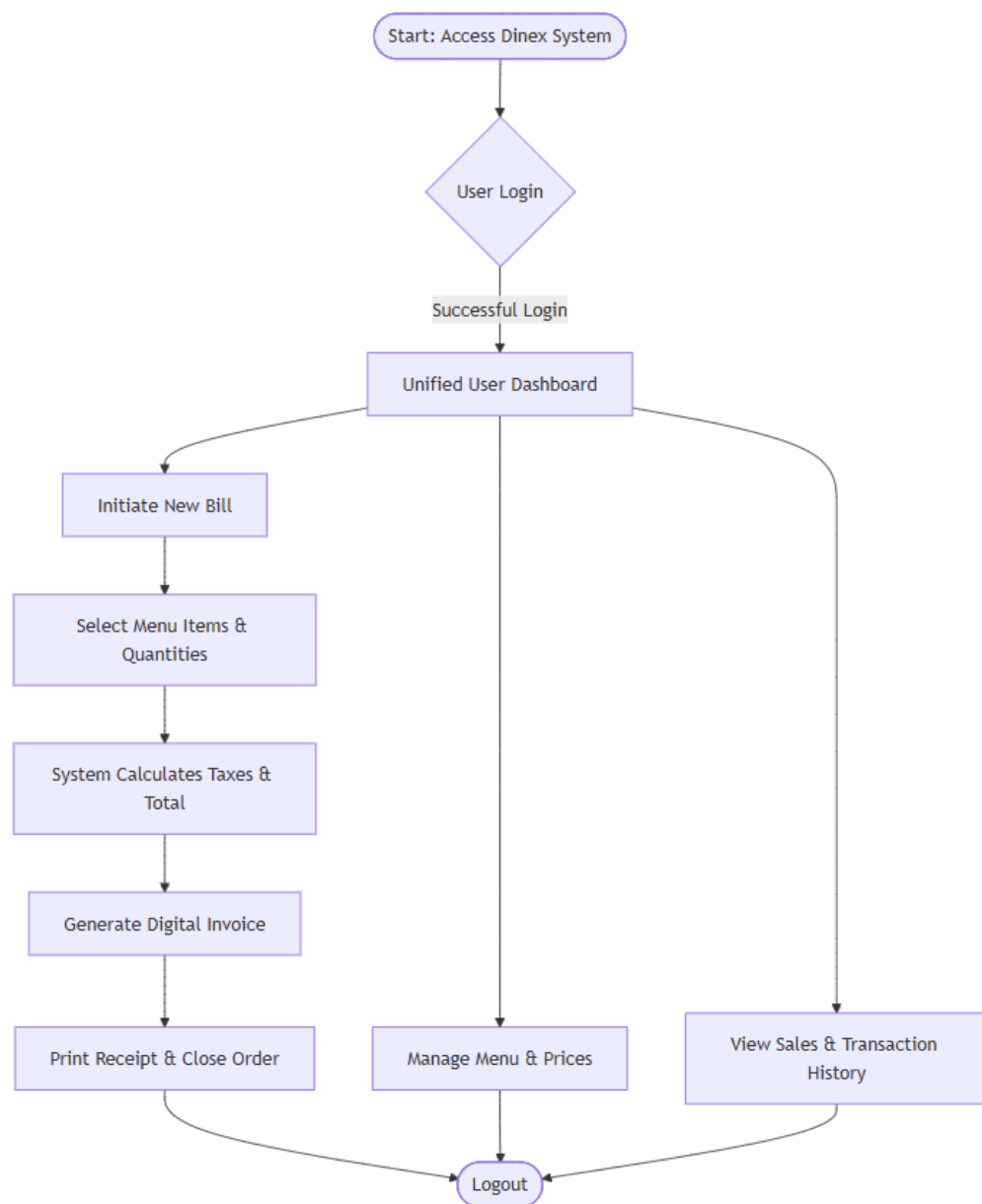
Daily Activity Report

- **Day 1:- 15th July 2026**

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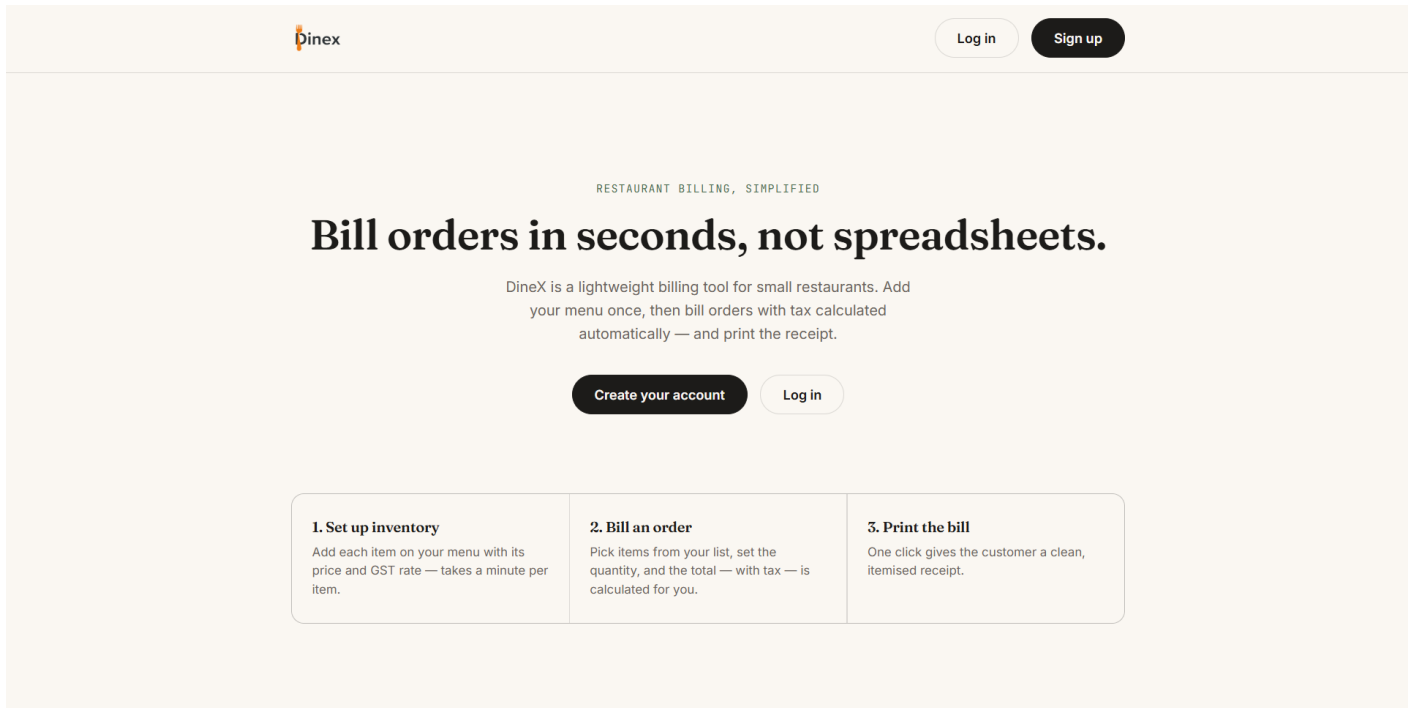
Made a team of 6 students, discussed the real-world problems and decided the problem statement for the web application and got approved by the guide. Finalized the Dinex billing and management web application.

- **Day 2:- 16th July 2026**
- **Designing of website architecture and user flow diagrams for the billing process.**



Day 3:- 17th July 2026

Designed the home page



Day 4:- 18th July 2026

Designed the login page

Dinex

Log in Sign up

Dinex

Log in

Welcome back — log in to your dashboard.

Username

Password

Log in

New to DineX? [Create an account](#)



- Day 5: 19th July 2026
- Sign up page

DineX

Log in Sign up

DineX

Create your account
Set up DineX for your restaurant.

Restaurant / your name

Username

Password

Create account

Already have an account? [Log in](#)

Dashboard



[Home](#)[Inventory](#)[Billing](#)

Hi, **Spice Factory**[Log out](#)

WELCOME BACK

Bill orders in seconds, not spreadsheets.

Add your menu once, then bill orders with tax calculated automatically — and print the receipt.

[Manage inventory](#)[Start billing](#)

1. Set up inventory

Add each item on your menu with its price and GST rate — takes a minute per item.

2. Bill an order

Pick items from your list, set the quantity, and the total — with tax — is calculated for you.

3. Print the bill

One click gives the customer a clean, itemised receipt.

Day 7: 21th July 2026

Designed the Menu Management (Add/Edit Prices) page



HomeInventoryBilling

Hi, Spice FactoryLog out

Inventory

Add the products your restaurant sells, with price and GST rate.

Product name

Price (₹)

GST (%)

Add product

PRODUCT	PRICE	GST %	
Veg Platter	₹860.00	5%	Remove
Paneer Dum Tikka	₹420.00	5%	Remove
Paneer Cotton Kabab	₹420.00	5%	Remove
Cheese Cherry Pineapple	₹380.00	5%	Remove
Paneer Rajwadi	₹360.00	5%	Remove
Shahi Paneer	₹360.00	5%	Remove
Panner Mussallam	₹380.00	5%	Remove
Chicken Malwani Bone-less	₹999.00	5%	Remove
Mutton Nawabi Handi	₹1,280.00	5%	Remove

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- Day 8:22nd July 2026
- Designed the core Billing Module for staff to select items, calculate totals, and generate invoices.



HomeInventoryBilling

Hi, Spice FactoryLog out

New Bill

Tap a product to add it to the current bill.

Butter Chicken
₹750.00 · GST 5%

Add

Cheese Cherry Pineapple
₹380.00 · GST 5%

Add

Chicken Angara
₹749.00 · GST 5%

Add

Chicken Malwani Bone-less
₹999.00 · GST 5%

Add

Mutton Biryani
₹849.00 · GST 5%

Add

Mutton Nawabi Handi
₹1,280.00 · GST 5%

Add

Paneer Cotton Kabab
₹300.00 · GST 5%

Add

Table Bill

Butter Chicken

- 1 +

₹787.50
₹750.00 × 1 · GST 5%

Cheese Cherry Pineapple

- 1 +

₹399.00
₹380.00 × 1 · GST 5%

Subtotal₹1,130.00

GST₹56.50

Total₹1,186.50

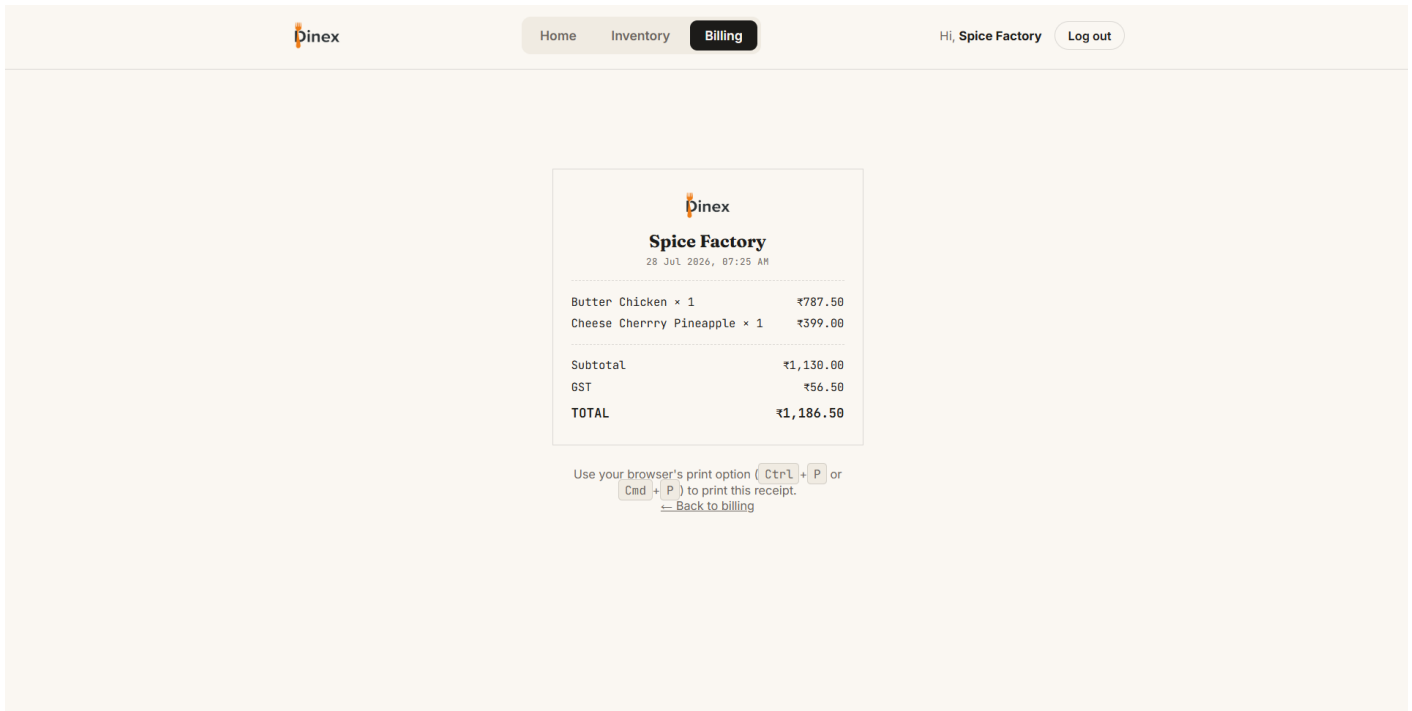
Print bill

Clear bill

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- Day 9: 23rd July 2026
- Designed the receipt



- **Day 10: 24th July 2026**
- **Performed unit and cross testing**

- **Day 11: 25th July 2026**
- **Deployment of the website**

Result

The Dinex Web Application was successfully developed and tested.

The system provides:

- Secure Staff/Admin Login
 - Dynamic Menu & Price Management
 - Automated Billing & Tax Calculation
 - Admin Dashboard for Sales Overview
 - Transaction History Tracking
 - Database Management
 - Responsive User Interface
- The application successfully reduces manual work and improves the accuracy and efficiency of processing restaurant bills.

Conclusion

The Dinex Web Application successfully provides a secure, efficient, and user-friendly solution for managing billing and financial records within a restaurant setting. The application simplifies the checkout process, enables accurate revenue tracking, reduces manual calculation errors, and improves operational transparency.

This project also enhanced our practical knowledge of web development, database management, version control using Git & GitHub, and collaborative software development.

List of References

- PHP Documentation — <https://www.php.net>
- MySQL Documentation — <https://dev.mysql.com>
- MDN Web Docs — <https://developer.mozilla.org>
- W3Schools — <https://www.w3schools.com>
- Bootstrap Documentation — <https://getbootstrap.com>
- Font Awesome — <https://fontawesome.com>
- Git Documentation — <https://git-scm.com>
- GitHub — <https://github.com>
- XAMPP — <https://www.apachefriends.org>
- Visual Studio Code — <https://code.visualstudio.com>